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**GIS AND REMOTE SENSING**  
**MICROPROJECT**

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# Spatial Analysis of Solar Power Generation Potential in Maharashtra Using QGIS

# **1. Abstract**

The growing emphasis on renewable energy as a sustainable solution to environmental degradation and fossil fuel dependence has highlighted the importance of solar energy, particularly in sun-rich regions like Maharashtra, India. This research utilizes Geographic Information System (GIS) techniques to evaluate the potential for solar energy generation across Maharashtra by analyzing multiple spatial factors. Key environmental datasets, including Global Horizontal Irradiance (GHI), cloud cover, temperature, and land use/land cover (LULC), were collected and processed using QGIS software. Each dataset was reclassified based on its suitability for solar power production. The study generated individual suitability maps for each factor, which will later be integrated into a weighted overlay analysis to produce a comprehensive solar suitability map. This approach aims to provide spatial decision-makers with critical insights into optimal locations for solar farm development, taking into account environmental compatibility and resource efficiency.

## **2. Introduction**

As the world moves toward sustainable energy solutions, solar power has emerged as a key alternative to fossil fuels, particularly in sun-rich countries like India. Maharashtra, with its high solar insolation and growing energy demand, presents strong potential for solar energy development. Identifying optimal solar sites, however, requires the integration of multiple climatic factors.

GIS provides a powerful tool for analyzing spatial data and supporting decision-making in renewable energy planning. This study uses GIS techniques to assess solar potential in Maharashtra by examining three critical parameters: Global Horizontal Irradiance (GHI), cloud cover, and temperature. Each factor is processed, reclassified, and visualized to create suitability maps, forming the foundation for a composite solar suitability assessment. This research aims to support data-driven planning for efficient solar energy utilization across the state.

## **3. Objectives**

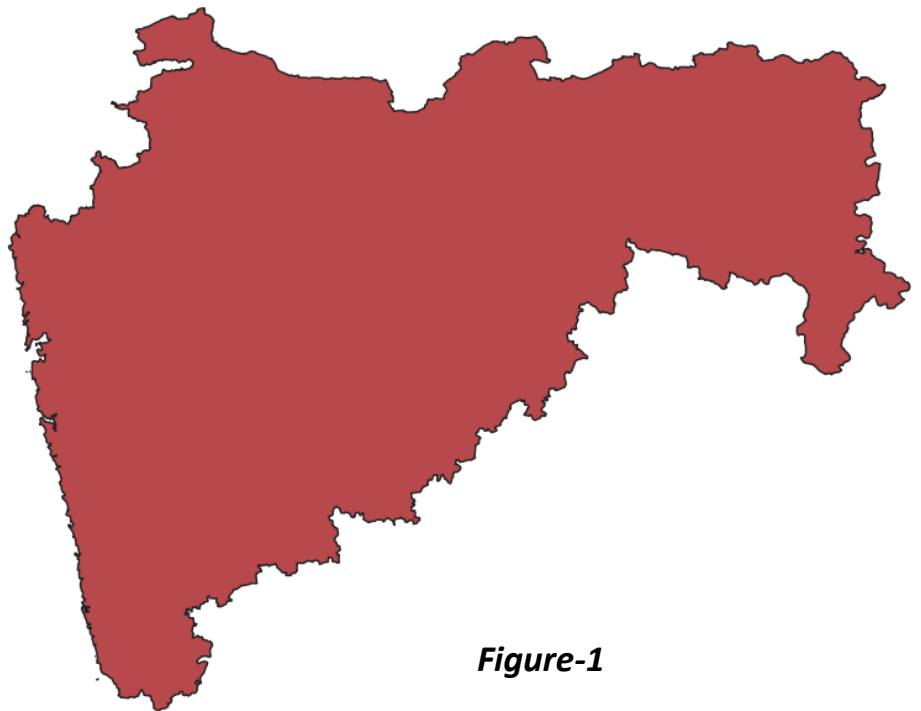
- To analyze solar radiation, cloud cover, temperature, and LULC data using GIS techniques.
- To reclassify and score each factor based on suitability for solar energy generation.
- To prepare suitability maps for each factor.

## 4. Study Area

Maharashtra, located in western India, is the second-most populous state and one of the most industrially developed regions of the country. Geographically, it spans between latitudes **15.6°N to 22°N** and longitudes **72°E to 80.9°E**, covering an area of approximately **307,713** square kilometers. The state's topography is diverse, **ranging from the low-lying Konkan coastal plains to the elevated Western Ghats and the Deccan Plateau**. This geographical variety leads to significant climatic and environmental variation, which directly affects solar energy potential across different parts of the state.

Maharashtra was selected as the focus for this solar energy assessment project for several strategic reasons. **First, the state receives high**

**solar irradiance year-round, especially in its central and eastern districts, making it naturally favorable for solar power generation. Second, Maharashtra's rapidly growing**



*Figure-1*

**energy demand due to industrialization and urban expansion necessitates the exploration of sustainable energy alternatives.** Moreover, the Government of Maharashtra has launched various initiatives, such as the "**Maharashtra Renewable Energy Policy**," which encourages solar power adoption through subsidies and investment incentives. These factors make the state not only suitable but also strategically important for large-scale solar energy deployment.

## **5. Parameters**

In order to evaluate the solar potential across Maharashtra comprehensively, the following four key spatial-environmental parameters were selected:

- **Global Horizontal Irradiance (GHI):** This represents the amount of solar radiation received per unit area. It is the most critical factor, as higher GHI directly translates to more solar energy available for conversion.
- **Cloud Cover:** Cloudiness reduces the amount of direct solar radiation reaching the surface. Areas with frequent cloud cover are less ideal for solar panel installation. Thus, this parameter helps in filtering out regions with less reliable sunlight availability.
- **Temperature:** While sunlight is essential, extremely high ambient temperatures can negatively impact the efficiency

of photovoltaic panels. Therefore, areas with moderate, stable temperatures are preferred for consistent and optimal energy output.

## **6. Data and Tools Used**

Software Platform:

- **QGIS (Open-source GIS software)**

Spatial Datasets and Sources:

- **Solar Radiation (GHI): CSV raster data from NASA POWER**
- **Cloud Cover: CSV raster data from NASA POWER**
- **Temperature: CSV raster data from NASA POWER**
- **Maharashtra Boundary: Shapefile from GADM**

These datasets were chosen for their direct influence on the efficiency of solar photovoltaic systems. GHI reflects available solar energy, cloud cover affects sunlight availability, and temperature influences panel performance and longevity.

## 7. Methodology

### 7.1 Data Collection

Environmental datasets were collected from open-source platforms in CSV format and converted into raster layers using QGIS. Each dataset was checked for completeness and accuracy.

### 7.2 Data Preprocessing

- All raster layers were projected uniformly to ensure consistency.
- Layers were clipped to the Maharashtra boundary using the shapefile.
- Data was analyzed over two periods: six months of summer and six months of winter to account for seasonal variation.

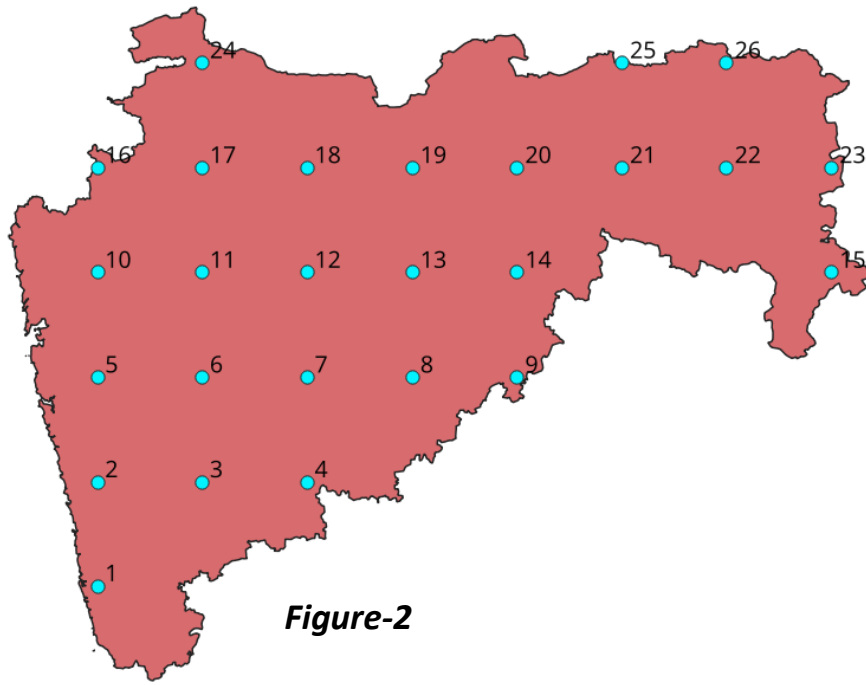
### 7.3 Factor-wise Reclassification

Each environmental factor was reclassified to reflect its influence on solar potential:

- **GHI:** Higher values scored higher (scale 1–5).
- **Cloud Cover:** Lower cloud percentages were more favorable.
- **Temperature:** Optimal ranges (around 25–26°C) received higher scores. Symbology was applied using the singleband pseudocolor scheme to visually interpret suitability levels.



## 8. Analysis



**Figure-2**

To enable a more structured and localized analysis of solar energy potential across Maharashtra, the state has been spatially divided into **26 distinct sectors**.

This division is based on a grid overlay aligned with the spatial distribution of

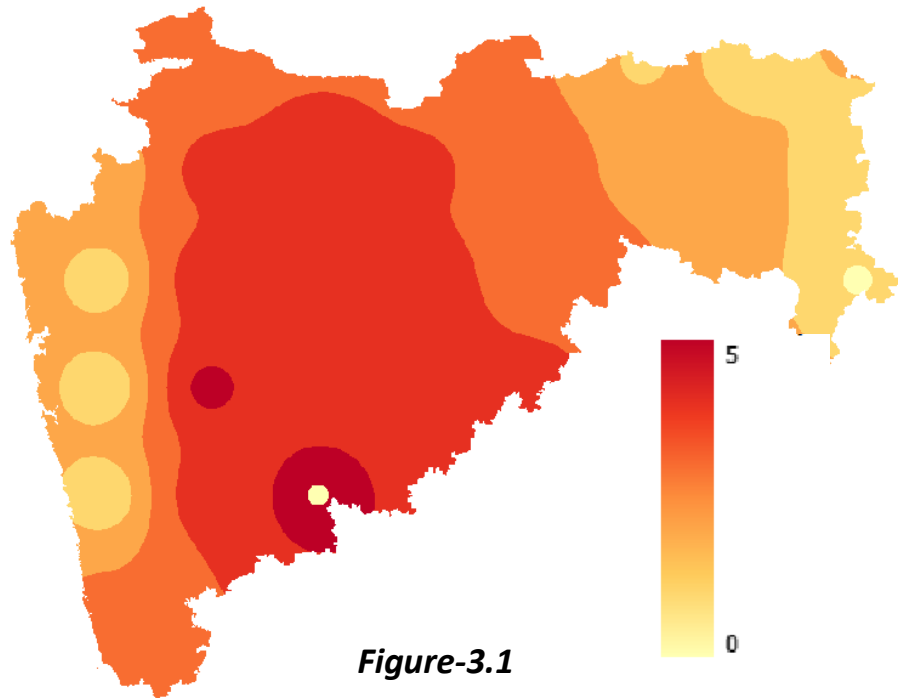
the key parameters—**Global Horizontal Irradiance (GHI)**, **cloud cover**, and **temperature**—that influence solar energy efficiency.

Each sector is labeled numerically (e.g., *Pt-2* or *Point 2*) for ease of reference in the analysis and mapping stages. As shown in **Figure 2**, these sectoral divisions facilitate targeted evaluation by allowing the comparison of environmental conditions across various parts of the state.

Subsequent analysis and interpretations will be conducted sector-wise to determine which specific areas demonstrate the most favorable conditions for hosting **ideal solar power facilities**.

## 8.1 GHI INDEX:

March-September (2022)  
(Summer)



*Figure-3.1*

The figure 3.1 above illustrates the **Global Horizontal Irradiance (GHI) distribution across Maharashtra for the months of March to September, which broadly represents the summer season in the region.** This period is critical for solar energy assessments as it coincides with the peak solar irradiance due to India's geographical position—**lying north of the equator and near the Tropic of Cancer—which results in higher sun angles and longer daylight hours during summer.**

In the visualization, **darker shades represent higher GHI values,** indicating areas with greater solar energy potential, while lighter shades correspond to lower GHI concentrations.

From the spatial distribution, it is observed that the central region of Maharashtra exhibits the **highest levels of solar**

irradiance during this period, making it highly favorable for solar energy applications.

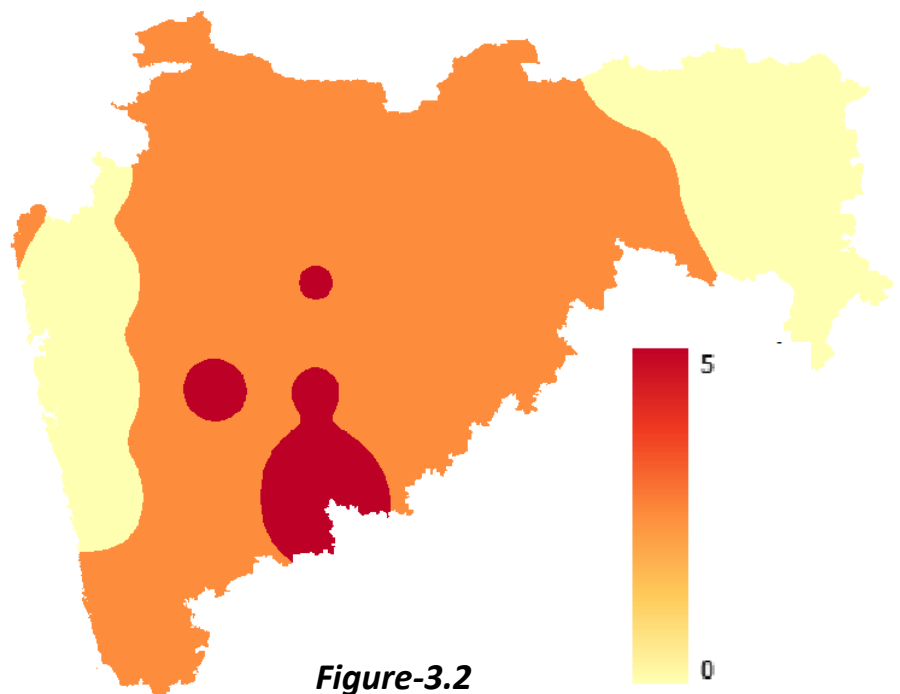
Based on the sectoral division, the following regions (sectors) recorded the highest GHI concentrations during the summer season:

1. **Point-4** (17.5°N,75.5°E)
2. **Point-6** (18.5°N,74.5°E)
3. **Point-12** (19.5°N,75.5°E)

These sectors demonstrate the most promising solar energy potential during summer and may serve as optimal locations for future solar infrastructure development.

**October-February (2022-23) (Winter)**

The figure presented above displays the **Global Horizontal Irradiance (GHI)** distribution across Maharashtra for the months of October to



February, encompassing the winter season. During this time, **solar irradiance tends to decrease compared to summer due to the sun's lower angle in the sky and shorter daylight hours.** However, regions near the equator, such as Maharashtra, **still receive considerable sunlight, making winter an important season for year-round solar energy evaluation.**

In the map, darker areas indicate higher GHI levels, while lighter regions reflect lower irradiance. **This visual classification assists in identifying zones that maintain significant solar potential even during the less intense winter period.**

The analysis reveals that central and southeastern Maharashtra continue to **exhibit relatively high GHI values** in winter, underlining their consistency as viable zones for solar power deployment throughout the year.

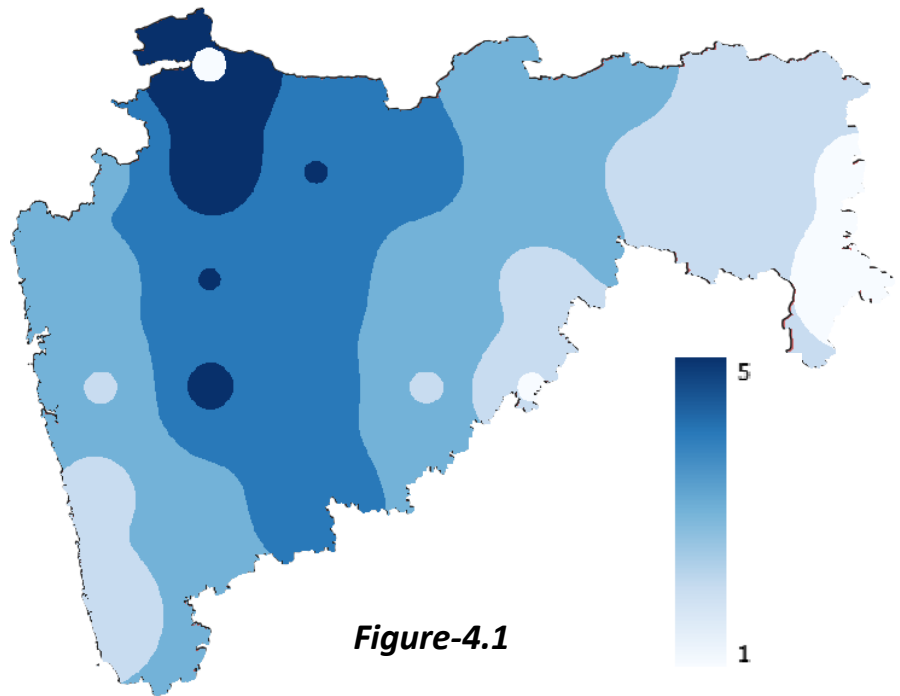
The sectors demonstrating the highest GHI concentrations during the winter season are:

- 1. Point-4** (17.5°N,75.5°E)
- 2. Point-6** (18.5°N,74.5°E)
- 3. Point-7** (18.5°N,75.5°E)
- 4. Point-12** (19.5°N,75.5°E)

These findings confirm that these zones remain strong candidates for solar energy generation, even during seasonal solar variation, reinforcing their suitability for sustained solar infrastructure.

## 8.2 CLOUD COVER:

March-September (2022)  
(Summer)



*Figure-4.1*

The figure-4.1 above presents the cloud cover distribution across Maharashtra during the summer season, **covering the months from March to September**. Cloud cover is a critical parameter in solar energy planning, as **high cloud density can significantly reduce the amount of solar radiation reaching the surface**, thereby affecting the efficiency and energy output of photovoltaic systems.

In the visual representation, **darker shades correspond to higher cloud cover, indicating less favorable conditions for solar generation, while lighter shades denote clearer skies with higher potential for solar energy capture.**

The extreme eastern part of the state is to be observed with lowest of the cloud covers in these part of the month.

**The same could also be said to the southern-western coast of the state. Moreover, the expanse and the rate of change of cloud cover is quite high, thanks to the proximity to the sea due to which the wind currents are always in motion**

**While central highlighted areas lack in clear sky.**

Analysis of the spatial pattern reveals that the central and western regions of Maharashtra exhibit relatively low cloud cover during this period, offering prolonged sunshine hours ideal for solar power production.

Based on the sectoral division, the sectors with the lowest cloud cover—and therefore the most favorable conditions—during the summer season are:

**1. Point 1**

**2. Point 2**

**3. Point 22**

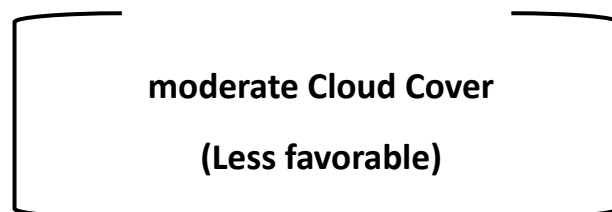
**4. Point 23**



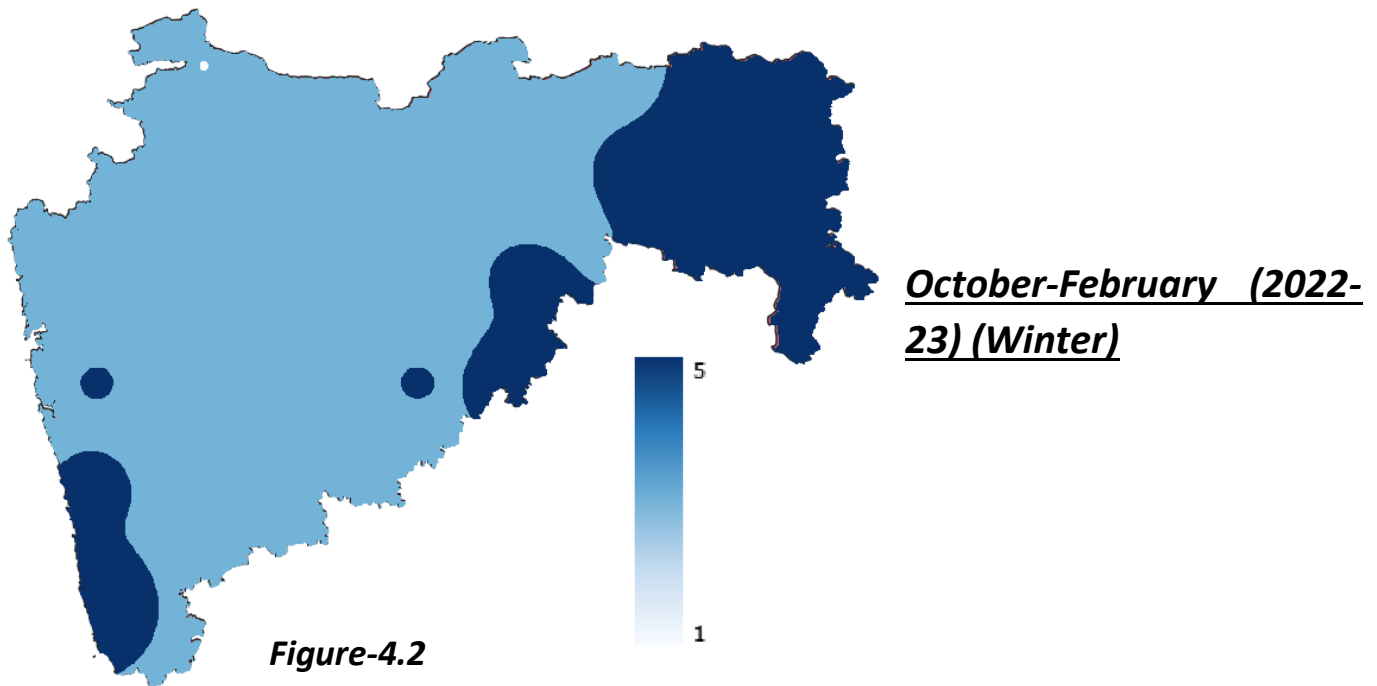
**5. Point 3**

**6. Point 7**

**7. Point 4**



These sectors are well-positioned for efficient solar panel performance during summer, owing to minimal atmospheric interference and maximum sunlight exposure.



The **Figure-4.2** above illustrates the **cloud cover distribution** across the state of **Maharashtra during the winter season**, spanning from **October-February**. As a parameter crucial to solar energy planning, cloud cover directly affects the availability of solar irradiance, with **increased cloud presence leading to reduced solar efficiency** and diminished photovoltaic output.

In this representation, also like the previous case, darker **shades indicate higher cloud concentration**, signifying areas that are less suitable for solar installations due to frequent sky obstruction. **Lighter shades denote clearer skies**, favorable for uninterrupted solar energy generation.

Contrary to the summer trend, **winter conditions demonstrate a significant shift in cloud cover patterns**. The **central and western sectors**, which previously exhibited high cloud density, now experience **lower cloud accumulation**. This seasonal variation can be attributed to changing atmospheric dynamics and shifting wind patterns during the winter months. Emerging as comparatively better-suited zones for solar energy generation in this season. The **inland positioning** and reduced maritime influence in these areas contribute to more stable and clearer skies.

Meanwhile, the **eastern and parts of the southeastern regions** of Maharashtra now exhibit **notably higher cloud cover**, emerging as comparatively better-suited zones for solar energy generation in this season. The **inland positioning** and reduced maritime influence in these areas contribute to more stable and clearer skies.

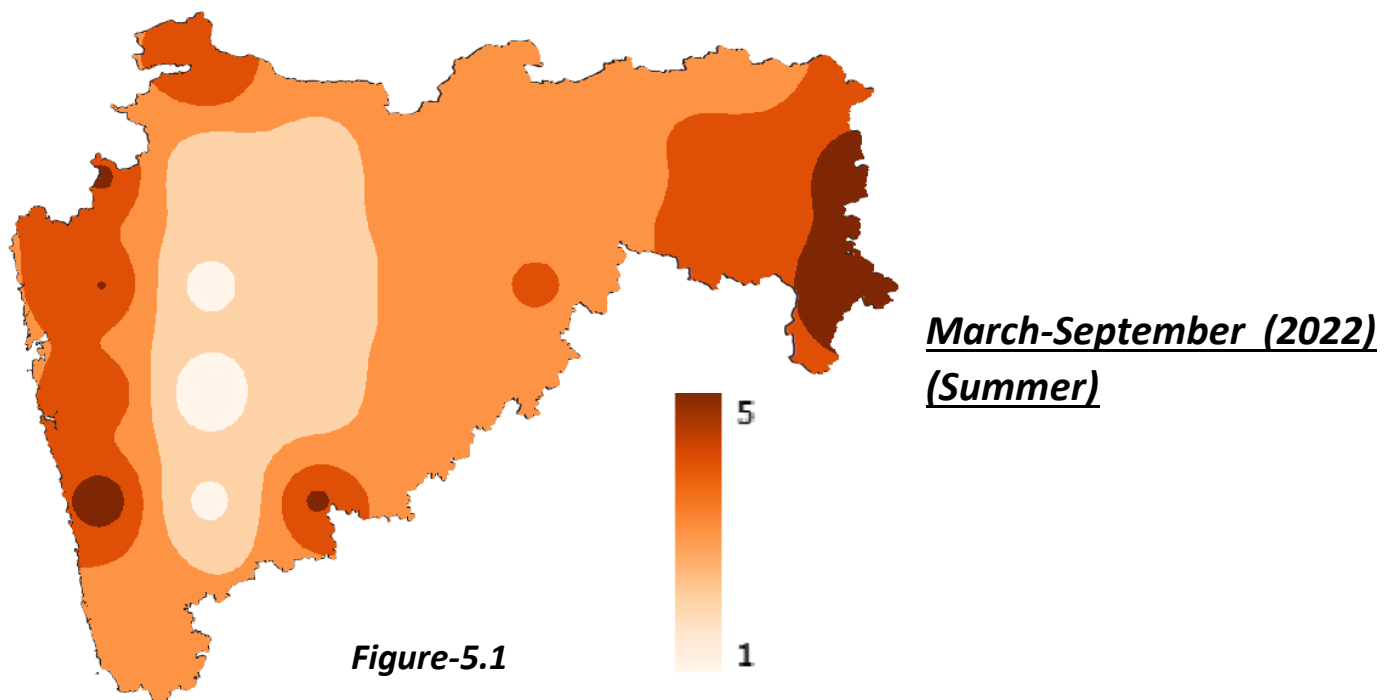
According to the sector-based classification, the following zones show **minimal cloud cover during winter** and are thereby most favorable for solar deployment:

1. **Point-6** (18.5°N,74.5°E)
2. **Point-7** (18.5°N,75.5°E)
3. **Point-3** (17.5°N,74.5°E)
4. **Point-4** (17.5°N,75.5°E)



These regions are identified as having **lower atmospheric obstruction** during winter, offering consistent exposure to sunlight and retaining solar potential even in months of winter.

### 8.3 Temperature:



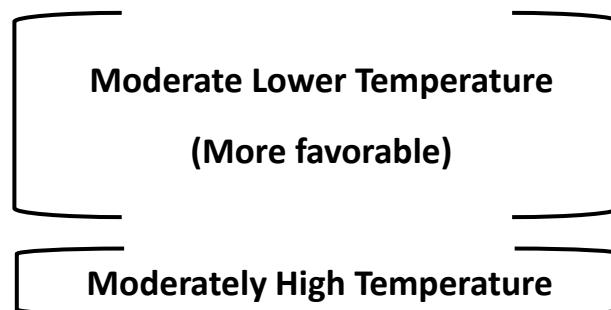
**Figure-5.1** presents the temperature distribution across Maharashtra during the summer season, extending from **March to September**. Temperature, while essential to **solar photovoltaic (PV) performance**, becomes a double-edged sword during this period. While ample sunlight is available, excessively high **temperatures can reduce the efficiency** of PV systems by increasing the resistance in solar cells, **leading to thermal losses**.

In the temperature raster map, lighter shades represent areas with moderate summer temperatures, which support optimal PV functioning. Conversely, darker shades indicate regions experiencing higher temperatures, which may lead to marginal declines in system output due to overheating.

During the summer months, **the northern, central, and southeastern parts of Maharashtra witness some of the highest temperature values**, sometimes exceeding the efficient operating range for standard PV systems. On the other hand, **certain western coastal and elevated regions benefit from slightly moderated temperatures, due to coastal breezes and topographic relief.**

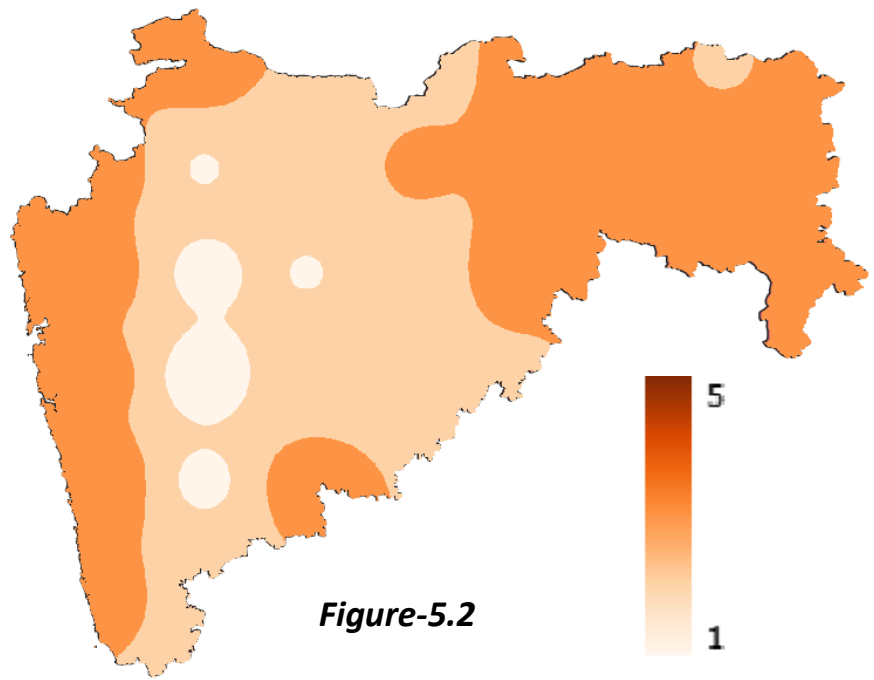
Upon examining the sector-based division of the state, the following points emerge as holding the most favorable temperature conditions for solar generation during the summer period:

- 1. Point 3**
- 2. Point 6**
- 3. Point 7**
- 4. Point 4**



These areas strike a balance between sufficient irradiance and manageable temperature levels, making them well-suited for efficient solar energy capture during the peak summer season.

October-February (2022-23)  
(Winter)



**Figure-5.2** illustrates the **temperature distribution across Maharashtra during the winter season**, encompassing the months from **October-February**. Temperature plays a vital role in the performance of solar photovoltaic (PV) systems. While sunlight is the primary source of energy, excessive heat can reduce the efficiency of solar panels, making **moderate temperature zones more desirable** for optimal energy conversion.

In this raster visualization, **lighter shades represent areas with moderate, optimal temperatures** (around 24–26°C), which are ideal for the functioning of most solar panels. In contrast, **darker shades indicate regions with lower or sub-optimal temperatures**, which may still permit solar generation but with reduced operational efficiency due to increased resistance and altered photovoltaic behavior.

During the winter months, the **temperature distribution across Maharashtra is more balanced**, with **central and eastern regions** generally maintaining a favorable temperature range. The **northern and high-altitude areas**, however, display relatively cooler temperatures that may slightly lower system efficiency.

Based on the division of the state into predefined sectors, the following points are observed to have **the most optimal temperature conditions** for solar facility planning during the winter period:

|                    |                                                                                      |
|--------------------|--------------------------------------------------------------------------------------|
| 1. <b>Point-6</b>  |    |
| 2. <b>Point-11</b> |                                                                                      |
| 3. <b>Point-7</b>  |   |
| 4. <b>Point-12</b> |                                                                                      |
| 5. <b>Point-4</b>  |  |

These sectors maintain **temperatures close to the ideal operational range**, supporting consistent and efficient solar power generation even in cooler months.

## 9. Findings

### 9.1 GHI Analysis

- During **summer (March–September)**, sectors such as **Point-4, Point-6, and Point-12** showed the highest solar irradiance values, marking them as prime zones for solar energy capture.
- In **winter (October–February)**, **Point-4, Point-6, Point-7, and Point-12** continued to exhibit high GHI values, confirming their year-round potential for photovoltaic deployment.

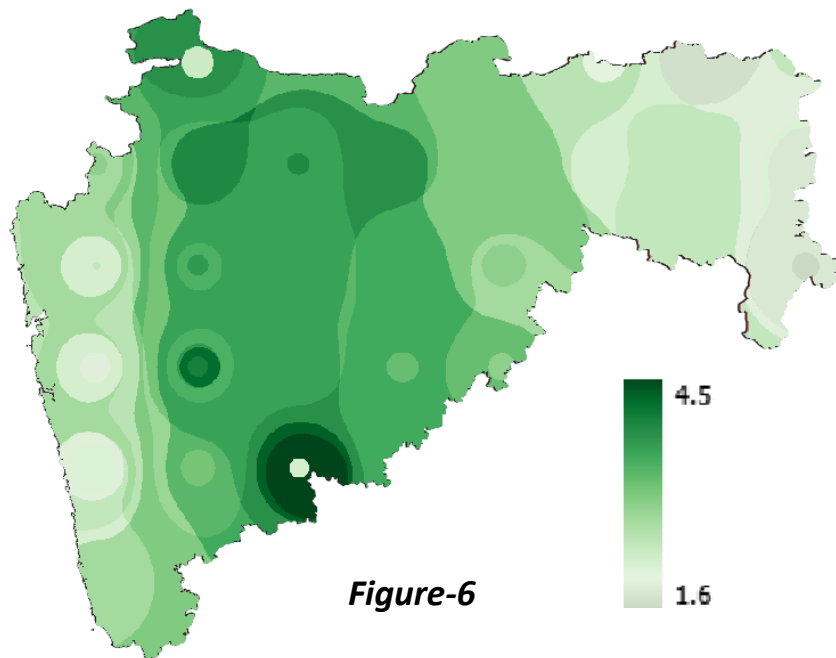
### 9.2 Cloud Cover Analysis

- In the **summer season**, minimal cloud interference was observed in **Point-1, Point-2, Point-22, and Point-23**, while **Point-3, Point-7, and Point-4** also displayed relatively low cloud cover.
- **Winter analysis** revealed a shift, where **Point-6, Point-7, Point-3, and Point-4** experienced the clearest skies, offering reliable solar radiation with minimal atmospheric obstruction.

### 9.3 Temperature Suitability

- During **summer**, **Point-3, Point-6, Point-7, and Point-4** maintained moderate to moderately high temperature levels—optimal for solar panel efficiency without significant thermal losses.

- In **winter**, sectors such as **Point-6, Point-11, Point-7, Point-12, and Point-4** maintained a balanced temperature range ( $\sim 24\text{--}26^{\circ}\text{C}$ ), aligning well with the operational threshold of standard PV systems.
- **Weighted Overlay Analysis:**  
A composite **Solar Suitability Index** was proposed using weighted contributions from each factor:



- The weights reflect the critical importance of **solar irradiance**, while also giving significant consideration to atmospheric and climatic consistency.
- The preliminary overlay highlights sectors like **Point 4, Point 6, Point 12, and Point 7** as **optimal zones** that

repeatedly perform well across all parameters.

- Conversely, **regions with consistently high cloud cover and temperature extremes** were deprioritized in the final suitability output.

These findings offer a **data-driven pathway for identifying optimal solar farm locations**, supporting sustainable energy planning and infrastructure development in Maharashtra.

| Sector<br>(Point<br>No.) | GHI<br>Performance | Cloud Cover<br>Suitability | Temperature<br>Suitability | Seasonal<br>Consistency | Overall<br>Feasibility |
|--------------------------|--------------------|----------------------------|----------------------------|-------------------------|------------------------|
| Point-4                  | High               | Low cloud presence         | Optimal                    | Consistent              | Very High              |
| Point-6                  | High               | Low cloud presence         | Optimal                    | Consistent              | Very High              |
| Point-12                 | High               | Moderate cloud presence    | Optimal                    | Consistent              | High                   |
| Point-7                  | High               | Low cloud presence         | Optimal                    | Consistent              | Very High              |
| Point-3                  | Moderate           | Low cloud presence         | Optimal                    | Partial consistency     | Moderate to High       |
| Point-1                  | Moderate           | Low cloud presence         | Moderate                   | Partial consistency     | Moderate               |
| Point-11                 | Moderate           | Moderate cloud presence    | Optimal                    | Inconsistent            | Moderate               |
| Point-9                  | Moderate           | Moderate cloud presence    | Moderate                   | Inconsistent            | Low to Moderate        |

### Summary:

- **Point-4** (17.5°N,75.5°E), **Point-6** (18.5°N,74.5°E), **Point-7** (18.5°N,75.5°E), and **Point-12** (19.5°N,75.5°E) demonstrate optimal performance across most parameters and are best suited for utility-scale solar infrastructure.
- **Points 3 and 1** can be considered for supplementary or smaller-scale installations.
- The consistent spatial performance of top-ranking sectors supports their prioritization in future solar energy planning.

## 10. Conclusion

The integration of environmental and land-use factors using GIS has enabled the creation of factor-specific solar suitability maps for Maharashtra. These maps serve as a foundation for a composite suitability analysis. The project so far confirms that GIS is a powerful tool for spatial decision-making in renewable energy planning. The next phase involves performing a weighted overlay to finalize the solar suitability index map.

## 11. References

1. NASA POWER Data Portal (<https://power.larc.nasa.gov>)
2. NRSC/Bhuvan Portal (<https://bhuvan.nrsc.gov.in>)
3. GADM Boundary Data (<https://gadm.org>)
4. <https://earthexplorer.usgs.gov/>